Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/14/2022
PI Name:	Rosenblatt, Charles S. Ph.D.		
Project Title:	Microgravity Studies of Nanoparticles and Topological Defects in Liquid Crystal Thin Films		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/02/2016	**End Date:**	12/01/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	Liquid Crystal Facility NOTE: End date changed to 12/1/2022 per NSSC information (Ed., 1/26/22)		
Key Personnel Changes/Previous PI:	CoInvestigator Philip Taylor retired in 2019. To substitute for Taylor's theoretical expertise, we have collaborated with Professor Samo Kralj (University of Maribor, Slovenia) and Professor Hillel Aharoni (Weizmann Institute of Science, Israel).		
COI Name (Institution):	Lacaze, Emmanuelle Ph.D. (Université Pierre et Marie Curie, France) Hegman, Torsten Ph.D. (Kent State University) Lemieux, Robert Ph.D. (COLLABORATOR / University of Waterloo) Kralj, Samo (COLLABORATOR / University of Maribor, Slovenia) Aharoni, Hillel (COLLABORATOR / Weizmann Inst. of Science, Israel)		
Grant/Contract No.:	NNX17AC76G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	FINAL REPORT Original Description: Nanoparticle / liquid crystal composites have created new avenues for scientific discovery and technological development. For example, recent studies have shown that mixtures of liquid crystals and appropriate semiconducting nanoparticles can be exploited for potential high efficiency and easily manipulatable photovoltaic devices. But the development of this and other technologies requires a fundamental understanding of how the nanoparticles incorporate and move within the liquid crystal, how they aggregate in and around topological defects, and how the nanoparticles affect the defects themselves. We have shown that fluorescent carbon dots (FCDs) may be dispersed in liquid crystals and used as a probe of defects, demonstrating that they accumulate in regions of high liquid crystal curvature so as to relieve the local strain. As a result their emitted light can serve as a quantitative indicator for the interaction and evolution of topological defects. Interpretation of these initial results, however, is compromised by the presence of substrates, which is required in an Earth-bound environment. Here we perform experiments in the absence of a container by utilizing the Observation and Analysis of Smectic Islands In Space (OASIS) module aboard the International Space Station (ISS). By studying thin flat liquid crystal films doped with FCDs -- this is done by dissolving the FCDs in a solvent and "shooting" slowly moving pico droplets of the mixture at the liquid crystal film -- in a microgravity environment, we will obviate the need for bounding substrates, thereby rendering a much simpler system for interpretation. In particular, we will use OASIS' cameras to study the spatial distribution and transport of the FCDs in and around topological defects in the smectic-C phase of the liquid crystal, especially close to the smectic-A phase transition temperature. A related project will be undertaken in which the film will be bombarded by droplets of methanol solvent from a pico inkjet printer involving a large mechanical impulse. The impact of the droplets will induce the formation of "islands" and "divots" of smectic liquid crystal. The goal of this part of the project will be to determine the size and thickness of these induced structures as a function of the film thickness and the droplet speed. The work will involve both experimental and theoretical components. The results will provide a clear picture of how nanoparticles distribute themselves over time, how defects in the liquid crystal interact, and how islands form on a liquid crystal film. Added Description. With the termination of OASIS by NASA headquarters in 2021, the focus turned to the related topic of topological defects in liquid crystals. These occur if nanoparticles are shot at the film. Alternately, the defects can be patterned on a substrate using the stylus of an atomic force microscope. Here are sudying several phenomena, including the co-revolution of defects in an applied electric field and the role of defects on the topography of a liquid crystal film sitting on a substrate with a free (air) interface on the film's opposite side.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Liquid crystal (LC) textures and topological defects have been of interest for well over a century. Defects in two-dimensions generally are categorized in terms of the "defect strength"-- "s". Studies of defects can be used to obtain values of LC anchoring strength coefficients and elastic constants, albeit with limited accuracy. Defects are central in establishing the director profile in geometrically-confined liquid crystals, and serve as an important signature when determining the symmetry of phases, for example, distinguishing between a biaxial nematic phase and a Smectic-C (Sm-C) phase. Defect dynamics provide another important field of study, as defect motion is extremely sensitive to boundary effects and provides information about surfaces and impurities. Island and divot formation are useful in understanding the layer structure of the smectics and how layers reorganize. They provide information about the surface tension of the liquid crystal. Perhaps most important use of topological defects is that they can serve as traps for micro- and nanoparticles (NPs), as the presence of the particles reduces or eliminates the divergent energy associated with the defect core. In other words, nanoparticles (NPs) migrate within the LC in the direction of increasing stress associated with the liquid crystal's orientational order, as the presence of the particles provides relief from the large stresses that would be present at LC singularities. The incorporation of NPs into liquid crystals and their subsequent behavior in and around defects finds applications in areas such as energy harvesting, photonic structures, mirrorless lasing, micro-patterned LC devices, and chiral metamaterials for photonics and communications applications. They also may have applications in pharmaceuticals, as an alternative means of drug delivery.
Task Progress:	FINAL REPORT A patterned surface defect of integer "strength" can decompose into a pair of surface defects and disclination lines of half strength. For certain liquid crystals subjected to an applied AC electric field E, these half-integer defects are observed to wobble azimuthally for E > than some threshold field and, for sufficiently large fields, to co-revolve around a central point approximately midway between the two defects. This behavior is elucidated experimentally as a function of applied field strength E and frequency, where the threshold field for full co-revolution scales as the square root of frequency. Concurrently, nematic electrohydrodynamic instabilities were investigated. A complete field vs. frequency "phase diagram" compellingly suggests that the defects' fluctuations and eventual co-revolutions are coupled strongly to the presence of the electrohydrodynamic (EHD) instability. We have spent considerable effort to map the director field in the EHD instability and compare this to the director field of the defect pair in the absence of an electric field. As of this writing, we are at the cusp of showing that the EHD instability "carries" along the defect pair in its co-revolution motion. We also have examined the effects of chirality (the absence of mirror symmetry) on the motion, both in terms of mixing a chiral dopant into the liquid crystals (LC) and by patterning the substrate with a 2D chiral easy axis. A thin film of Smectic-A liquid crystal was deposited on a polymer-coated substrate that had been scribed with closely spaced (200 m separation) lines over a small square of side L, where L=85 μm. Because of the small size of the patterned region, material could be transported to form either a hill (for a thin film) or divot (for a thick film) that contains an "oily streak" LC texture above the scribed square. (The oily streak texture involves a series of half cylinders that are comprised of the smectic liquid crystal layers.) Optical measurements of the topography of the free surface of the film vs. average film thickness suggest that the oily streak layer attempts to adopt a preferred thickness that depends on the nature of the molecule, the temperature, and the surface tension at the air interface. We hypothesized that the oily streak has a preferred "Goldilocks" thickness z0: If the film is thinner than z0, liquid crystal material would flow into the patterned region to create a hill above the pattern; if the film is thick, then material would flow out of the region to create a divot. Competing against either hill or divot formation is surface tension, which attempts to keep the film surface flat. We developed a model in which we determine the energy cost of the oily streak layer if its thickness differs from the preferred thickness z0. Moreover, the model allows us to estimate the energy cost of the oily streak region for thickness deviations from z0. A manuscript has just been submitted for publication. We measured the behavior of a chirally-doped nematic mixture of of different helical pitches for a liquid crystal that aligns parallel to an applied electric field. Initially the liquid crystal molecules are "surface-stabilized" in the

	homeotropic configuration, where the molecules lie perpendicular to the cell walls.. By applying an in-plane magnetic field perpendicular to the director, we find a "Freedericksz" threshold field Hth above which the sample begins to undergo a combination of bend and twist distortions, and for which Hth decreases as the chiral helical pitch p decreases. For sufficiently tight pitch or sufficiently thick sample, the extrapolated Freedericksz threshold goes to zero. A similar experiment had been performed for a negative dielectric anisotropy LC using an electric field, but the current situation is of lower symmetry and therefore considerably more complex. We developed a theoretical model for this symmetry-broken configuration, finding good agreement with the experiment. Additionally, the new magnetic results are compared to the previous electric field 2*pi-degenerate experiments. As of this writing a manuscript is being prepared, which we expect to submit for publication in early 2023. Skyrmions are broken-symmetry excitations with particle-like characteristics and which are topologically protected. In practice, this configuration involves a series of concentric circles in which the liquid crystal molecules are perpendicular to the circles in the center, i.e., at r = 0. Moving radially outward, the polar angle increases to 90 degrees (in the plane of the circles), while the azimuthal component of the director is tangent to the circles, forming a clockwise or counterclockwise circumnavigation. Theoretically, a quasi-2D lattice of half-skyrmions can be the stable ground state for a chiral liquid crystal in a cell with homeotropic (vertical) boundary conditions. We have begun to study these structures by patterning a substrate for the skyrmion's director field and examining the resulting behavior (with and without applied electric field) using fluorescent confocal microscopy. Our goal is to investigate how the natural pitch of the chiral liquid crystal is affected by the cell thickness and the patterned pitch length. Additionally, we aim to understand defect formation when one substrate is patterned for half- or full-skyrmion director fields. That is, for a given chiral liquid crystal helical pitch, is the skyrmion stable over a range of spatial periodicities that can be imposed by a surface patterning technique? And if so, what is the nature of the required defect near the opposing substrate?
Bibliography Type:	Description: (Last Updated: 12/04/2024)
Articles in Other Journals or Periodicals	Harkai S, Cordoyiannis G, Susser AL, Murray BS, Ferris AJ, Rožic B, Kutnjak Z, Rosenblatt C, Kralj S. "Manipulation of line defect assemblies in plane-parallel nematic liquid crystals." Liquid Crystals Reviews. 2021 Nov 28. Submitted. https://doi.org/10.1080/21680396.2022.2042745 , Nov-2021
Articles in Peer-reviewed Journals	Harkai S, Cordoyiannis G, Susser AL, Murray BS, Ferris AJ, Rožič B, Kutnjak Z, Rosenblatt C, Kralj S. "Manipulation of mechanically nanopatterned line defect assemblies in plane-parallel nematic liquid crystals." Liquid Crystals Reviews. 2022 Mar 3;10(1-2):98-122. https://doi.org/10.1080/21680396.2022.2042745 , Mar-2022
Articles in Peer-reviewed Journals	Missaoui A, Susser AL, Aharoni H, Rosenblatt C. "Energetics of topographically designed Smectic-A oily streaks." Soft Matter. 2023 May 16;19(20):3733-8. http://dx.doi.org/10.1039/D3SM00306J , May-2023
Articles in Peer-reviewed Journals	Missaoui A, Susser AL, Aharoni H, Rosenblatt C. "Magnetic field-induced Freedericksz transition in a chiral liquid crystal." Applied Physics Letters. 2023 Mar 27;122(13). https://doi.org/10.1063/5.0146506 , Mar-2023